git diff command

-> The git diff is an informative command that shows the differences between two commits.

-> It is used to compare the changes made in one commit with the changes made in another commit.

-> Git consider the changed versions of same file as two different files. Then it gives names to these two files and shows the differences between them.

-> Common Uses of git diff:

1) Difference between working area and staging area.
   Syntax: git diff

2) Differences Between Staging Area and Last Commit
   Syntax:git diff --staged

3) Differences Between Two Commits:
   -> Syntax: git diff commit1 commit2

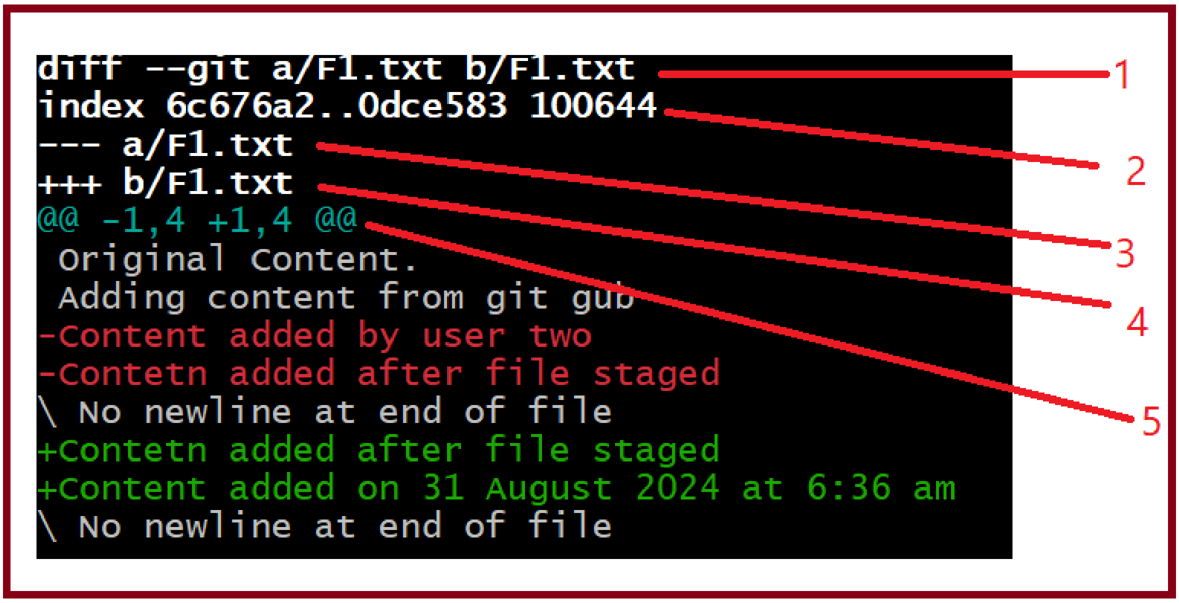
4) Difference between branches
   ->1) Your current with another local branch
      Syntax: git diff branch-name

   ->2) Your current branch with remote branch
      Syntax: git diff origin/remote-branch

   -> 3) Supposer your current branch is 'bug-fix-one' and you compare local main with remote then
      Syntax: git diff main origin/main

How to read the output of git diff command?

Consider the below output



Line 1 indicate file comparison header

a/F1.txt means F1.txt Before modification

b/F1.txt means F1.txt after Modification

Line 2 is a index line indicate git hashes of two version here 100644 is a file mode

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Line 3 : ---a/F1.txt (File Before Modification)

Line 4 : +++a/F1.txt (File After Modification)

@@ -1,4 : Original File start at line number 1 and affecting line 4

+1,4 : Modified File start at line number 1 and affecting line 4

Other symbols

- symbol indicate delettion or modication

+ indicate addition of modificatio of the line

git fetch command:

-> git fetch command is used to fetch the updates from remote repository.
-> git pull command is also used to fetch the updates from remote repository.

-> Working difference between git pull and git fetch is-> git pull command pull the changes from remote repository and merge these changes into your working directory automatically where as git fetch command fetch the updates from remote repository but does not merge them into your working directory automatically.

-> git fetch command is useful when you want to see what changes have been made to the remote repository without affecting your current branch or working directory.

-> git fetch command downloads the below things from remote repository.
1) latest commits,
2) Newly created branches
3) Delete branch.
3) Tags etc.

-> Usage of git fetch command

1) Fetch all updates from default Remote(Origin)
-> Syntax: git fetch

2) Fetch Updates from a Specific Remote
-> Syntax: git fetch origin

Above two commands are same but in second command explicitly we have specified remote name.

3) Fetches updates only for the specified branch
-> Syntax: git fetch origin branch-name

4) Fetche updates from all remotes you have configured in your repository.
-> Syntax: git fetch --all

5) Syntax:git fetch --prune

-> Fetch and Prune Deleted Branches

-> When branches are deleted on the remote repository, the references to these branches still exist in your local repository.
The --prune option removes these outdated references from your local repository.

Practical Implementation:

1. Create two new Folders (User1 and User 2) in your workspace folder (Devops)
2. Clone Same repository in User1 folder and User2 Folder
3. Open git bash in both repository seperatly
4. User1 will modify the file and push his changes on remote repo
5. User 2 will check the updates by using "git fetch" command.

Q. How to use "git fetch"?

-> First fetch remote changes by using "git fetch"

-> Check the changes by using "git log" or "git diff" or "git status"

-> After fetching, if you want to integrate the fetched changes into your local branch, you can either merge or rebase:

Merge Syntax: git merge origin/main

(If you are main branch and another user has push changes on main)

i.e if you want to merge other branch changes into your branch

then first switch to your branch and execute command

git merge target_branch

-> target_branch means From which branch your are going copy the changes

Rebase Syntax: git merge origin/main